

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1813

Lincoln 100 kV
October 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1813. The Interconnection Customer proposes a 177.3 MW solar plus storage facility interconnecting at the Lincoln 100 kV substation in the Illinois area. The request was received on July 13, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 578 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$31,320,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
E J STONEMAN STATION to MOORHEAD 100 KV #2	E J STONEMAN STA to MOORHEAD 100 KV Line	144.1	144	26.7
WILTON to LINCOLN 100 KV #2	LINCOLN to WILSON 100 KV Circuit 1	136.5	186	26.0
MOORHEAD to E J STONEMAN STATION 100 KV line	MOORHEAD to E J STONEMAN STA 100 KV line	134.8	154	26.7
E J STONEMAN STA to AIR PRODUCTS 100 KV #1	E J STONEMAN STA to AIR PRODUCTS SUBSTATION 100 KV Line 2	134.6	161	26.2
AIR PRODUCTS to WILTON 100 KV	AIR PRODUCTS to WILTON SUBSTATION 100 KV #1	132.0	178	25.7
LINCOLN to WILTON 100 KV Line 1	WILTON to LINCOLN 100 KV Circuit 2	131.5	193	26.0
E J STONEMAN STATION to AIR PRODUCTS 100 KV #2	AIR PRODUCTS to E J STONEMAN STA 100 KV	130.0	167	26.2
BRINGHURST to MOORHEAD 100 KV Cir. 1	BRINGHURST to MOORHEAD 100 KV Cir. 2	129.2	140	26.1
MOORHEAD to BRINGHURST 100 KV Cir. 2	BRINGHURST to MOORHEAD 100 KV Ckt 1	129.2	140	26.1
WILTON to AIR PRODUCTS 100 KV Ckt 1	AIR PRODUCTS to WILTON 100 KV	125.4	187	25.7
BRINGHURST to NAGUABO 2701 100 KV	GOODINGS GROVE to BRINGHURST 100 KV	118.8	105	24.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E J STONEMAN STATION to MOORHEAD 100 KV #2	Rebuild line with higher-capacity conductor	\$1,777,000	94% (\$1,679,000)
WILTON to LINOLN 100 KV #2	Rebuild line with higher-capacity conductor	\$2,475,000	100% (\$2,475,000)
MOORHEAD to E J STONEMAN STA 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$2,249,000	94% (\$2,125,000)
E J STONEMAN STA to AIR PRODUCTS 100 KV	Rebuild line with higher-capacity conductor	\$3,148,000	100% (\$3,148,000)
AIR PRODUCTS to WILTON 100 KV	Rebuild line with higher-capacity conductor	\$2,748,000	100% (\$2,748,000)
LINCOLN SUBSTATION to WILTON 100 KV #1	Rebuild line with higher-capacity conductor	\$3,407,000	100% (\$3,407,000)
E J STONEMAN STATION to AIR PRODUCTS 100 KV #2	Rebuild line with higher-capacity conductor	\$3,584,000	100% (\$3,584,000)
BRINGHURST to MOORHEAD 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$3,232,000	100% (\$3,232,000)
MOORHEAD to BRINGHURST 100 KV Cir. 2	Rebuild line with higher-capacity conductor	\$2,316,000	100% (\$2,316,000)
WILTON to AIR PRODUCTS 100 KV Ckt 1	Rebuild line with higher-capacity conductor	\$3,086,000	100% (\$3,086,000)
NAGUABO 2701 to BRINGHURST 100 KV line	Rebuild line with higher-capacity conductor	\$3,520,000	100% (\$3,520,000)
	Total allocated to project		\$31,320,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.